

LIAM MORLEY

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EDUCATION

Northwestern University | Weinberg College of Arts and Sciences

Evanston, Illinois

Bachelor of Arts | Major: Mathematics | Minor: Computer Science

September 2024 – June 2027

- **GPA: 3.72/4.00** | SAT: 1520/1600
- **Relevant Coursework:** Real Analysis, Mathematical Optimization, Data Structures and Algorithms, Probability and Stochastics, Combinatorics and Graph Theory, Mathematical Methods in Finance, Introduction to Artificial Intelligence

Boston College

Chestnut Hill, Massachusetts

Bachelor of Science | Mathematics and Computer Science

August 2023 – May 2024

- **GPA: 3.86/4.00**

PROFESSIONAL EXPERIENCE

Tidal Financial Group

Oak Brook, Illinois

Incoming Trading Analyst Intern

June 2026 – August 2026

- ETF Trading Intern involved with portfolio management and trade execution for active and index-based ETF funds

C3 Laboratories

Evanston, Illinois

Research Assistant

January 2025 – June 2025

- Contributed as a research assistant on the Satyrn Machine Learning Team with a specialty in Data Science Expansion
- Helped develop and implement new data analytic packages and systems for the Satyrn software using Apache Spark
- Implemented automatic regression and data cleaning models in Satyrn API

Italian Riviera

Waldwick, New Jersey

Deli Clerk

June 2022 – August 2024

- Kept store clean and organized while serving customers and handling monetary transactions
- Trained new employees on proper hygiene, customer service skills, and company policy
- Prepared sandwiches and other food items in a quick and efficient manner for customers

PERSONAL PROJECTS

Collegiate Rowing Ranking Model

- Analyzed historical collegiate rowing results to create predictive model based on predicted times and race margins with R^2 of 0.84
- Built RandomForestRegressor and Data Pipeline using SciKit Learn and Pandas to achieve error margin of ~ 4 seconds
- Utilized Monte Carlo Simulations and Google PageRank algorithm to create team rankings using model

European and American Options Pricer

- Developed options pricing models for both European and American options using common industry methods including Monte Carlo and Binomial Option Pricing Trees
- Utilized Numpy and Scipy to optimize computation time with vectorized operations
- Compared results with industry standard benchmarks and theoretical pricing models

Machine Learning Library

- Created a PyTorch style machine learning library from first principles in C++ containing automatic differentiation, matrix operations, and Stochastic Gradient Descent.
- Designed Modular Neural Network Architecture utilizing smart pointers for safe memory management, and the creation of a flexible API for creating MLPs and Deep Neural Networks using a variety of different loss and activation functions.

ACTIVITIES

Boston College Experimental Math and Machine Learning Lab

Chestnut Hill, Massachusetts

Lab Member

August 2023 – May 2024

- Served as one of lab's initial members by attending weekly discussions and recruiting new members
- Attended lectures by faculty members and researchers on recent developments in both mathematics and machine learning

ADDITIONAL

Skills: Python, C++, SQL, Asset Pricing, Data Structures and Algorithms, Mathematical Optimization, Excel, Claude Code

Awards: Top 5 in High School Class, National Merit Commendation, Top 20% Northwestern Crypto Trading Competition

Interests: Rowing, Golf, College Basketball, Skiing